We can also find the winding resistance by using the Kelvin bridge method.

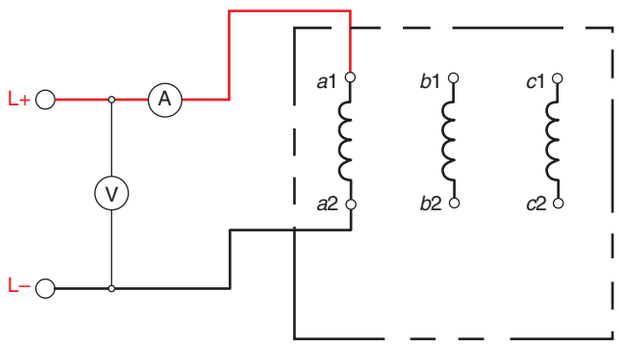


Fig. 10.3 Winding resistance test

2. Insulation resistance test

Megger is an instrument used for measuring high value of resistance and hence it is used to measure insulation resistance of winding. Two leads of megger are connected between winding terminal and the body of the motor. Now rotate the handle of the megger about 160 revolutions per minute (RPM).

Generally 1000v is used to measure the insulation resistance and the value should not be less than 1Mega ohm.

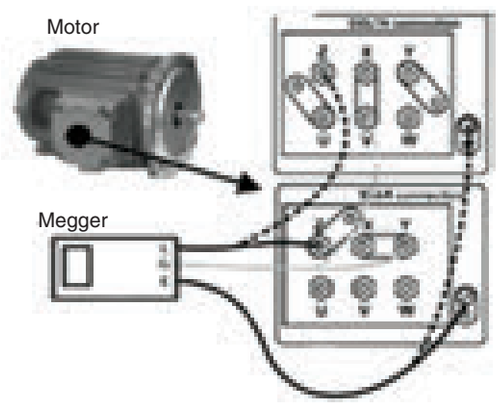


Fig. 10.4 Insulation resistance test

3. Growler test

Growler is an electromagnetic device which is used to find the open circuit, short circuit and ground faults in windings.

There are two types of growlers. They are

1. Internal growler
2. External growler

Internal growler is used to find faults in the stationary parts i.e, stator windings of motor, where as external growler is used to find faults in rotating parts i.e, rotor windings and armature windings of DC machines

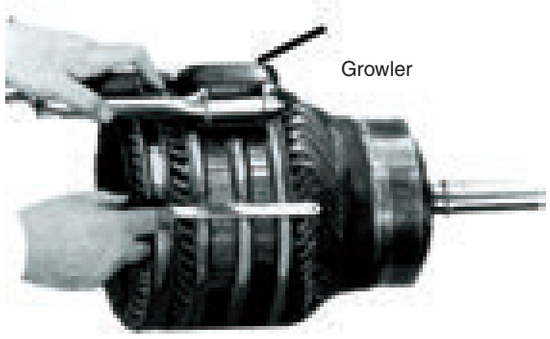


Fig. 10.5 Internal growler

Construction of growler

a. Internal growler

Internal growler consists of a winding over 'V' shaped laminated iron core. The legs were cut out on the top slantingly, to insert the legs into the stator slots easily. Internal growler has two types. They are

1. Internal growler with feeler
2. Internal growler without feeler

This type of growler with feeler is used in small size stators where we cannot insert growler to test the windings. In large type stators, there will be more space to test the winding and hence growler without feeler can be used. Here hacksaw blade is used instead of a feeler

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